Quantum Computing

Reading Guide

Compiled by the Spaced Research Ledger

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Quantum Computing is a survey of an industry at an awkward moment. The central scientific question of the last decade has been answered. The machines still cannot do anything commercially useful. Billions of dollars of public and private money moved into the field during 2026 anyway. This collection covers all three of those facts and the tension between them.

The answered question is error correction. Quantum computers are built from parts that fail constantly, and until recently nobody had shown that adding more parts made the whole thing better rather than worse. That threshold was crossed. What follows from it is an engineering programme measured in years, not a product.

The collection covers the field in ten parts. Parts 1 through 5 cover the competing hardware approaches, which are different physics rather than variations on a theme. Part 6 covers error correction, which every hardware approach depends on and which sets the price of everything else. Parts 7 through 10 cover the money, the theory and the claims.

Read front to back, the parts build on each other, since the error correction in Part 6 makes more sense once the hardware in Parts 1 through 5 is familiar. Read alone, every part stands on its own.

Every statement in every part carries a numbered citation to a public source, listed at the end of that part. When sources disagree, the disagreement is reported under its own heading rather than resolved by guesswork. Each part states when it was last updated.

How to Read This Collection

Each part opens with a summary written for someone who reads nothing else. Numbered sections follow, and each section starts with a short italic paragraph explaining in plain language what the thing is and why it matters. Inside each section, Recent Developments covers what changed lately and Current State covers what is durably true. Unresolved Conflicts, where present, records disagreements between sources with a note on what would settle each one.

Technical terms are explained once, on first appearance, and then used normally. Numbers are given in the form the sources used them, because in this field the exact figure usually is the finding.

Start with the summary of any part that touches your work. The intro of each part names its sections by number, so you can jump to what you need. Most readers should start with Part 6, which explains the constraint the whole field is organized around. Part 10 is the best antidote to a press release.

The Parts

Hardware by Modality

Five approaches compete here, and no two are close substitutes. They differ in what a qubit physically is, how long it survives, how fast it runs, and whether existing factories can build it. Each part reports the same measurements, so the parts can be read against each other.

Part 1 - Superconducting Qubits. The most widely deployed approach, behind most of the machines the public has heard of. Fast, manufacturable, and quick to lose its information, with IBM, Google, Rigetti, IQM, and a Fujitsu and RIKEN collaboration all pursuing different answers to the same trade.

Part 2 - Trapped Ion Qubits. Individual atoms held in electric fields, identical by nature and far more stable than any fabricated device, but a thousand times slower to operate. Holds the highest gate fidelities in the collection, and the largest logical-qubit counts.

Part 3 - Neutral Atom Qubits. Atoms held by focused laser light, packed thousands at a time where the other approaches count in hundreds. The scale is real and the atoms escape their traps, so most of the recent work is about replacing them faster than they leave.

Part 4 - Photonic Quantum Computing. Qubits made of light, needing no cooling and able to travel down ordinary optical fibre between racks. Photons get lost rather than corrupted, which is a different problem from every other part, and the largest machines here cannot be programmed for general work.

Part 5 - Spin Qubits. Single electrons in silicon, built on the same production lines that make ordinary computer chips. Behind on qubit count and ahead on manufacturing, this is the only approach an existing semiconductor industry could scale without inventing a new process.

Error Correction

Part 6 - Error Correction & Logical Qubits. How many unreliable physical qubits it takes to make one reliable logical qubit, and how fast that number is falling. Covers the result that proved the principle, the newer codes that cut the price by an order of magnitude, and the classical computing problem hiding inside the quantum one. Start here if you read only one part.

Industry, Academia & Claims

These three subjects are kept apart deliberately. Mixing them is what lets a press release read as a result.

Part 7 - Industry & Funding Moves. Who is paying, what governments have committed, which laboratories bought machines, and what companies have promised to deliver by when. Government money and public markets both now exceed private venture funding.

Part 8 - SPAC Mergers & Public Listings. Six companies went public between February and August 2026. Covers each deal's terms and what the companies have reported since. The earlier cohort that went public in 2021 delivered twelve cents on each projected revenue dollar.

Part 9 - Academic & Theoretical. New algorithms, proofs about what quantum computers can and cannot do faster, and the steady work of dequantization, which is showing that a claimed quantum speedup was never real. The negative results here are the most useful entries.

Part 10 - Applications & Benchmarking Claims. Every claim that a quantum computer did something useful, paired with the rebuttal it attracted. One supremacy claim in this part was answered by a laptop. As of the most recent assessment, no quantum advantage has been demonstrated for any real-world financial application.

About This Report

The Spaced Research Ledger is an automated system that gathers claims from live public sources, checks each one against its origin, and records every disagreement instead of merging it. Every stage runs on a different AI model, drawn from four to seven systems across competing labs, so no single model both finds a claim and judges it. The Spaced Research Ledger is designed and maintained by Ridley DiSiena.